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| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 12/02/2010 |
| **PI Name:** | Catauro, Patricia M.S. | | |
| **Project Title:** | Bulk Overwrap Packaging | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Space Human Factors Engineering | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SHFH**:Space Human Factors & Habitability (archival in 2017) | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 77058 | **Congressional District:** | 22 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 03/01/2010 | **End Date:** | 10/01/2010 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | Previous PI was Thomas Oziomek, from 10/01/2008 until 02/28/2010. See Oziomek for prior year reports. | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | Directed Research | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | The National Aeronautics and Space Administration (NASA) has established the goal of extending manned space exploration beyond low-earth orbit (LEO). Extended-duration missions of this type require considerable quantities of food to be flown into space, which presents two main challenges in designing a food system. The first is in maintaining the quality of the food throughout its shelf life (which may be in excess of five years), and the second is in assuring the mass and volume of the food system are optimized. The Bulk Overwrap Task has sought to identify a flexible bulk overwrap packaging system to meet these challenges, by providing a low mass food containment system that would prevent oxygen and moisture interaction with foods. Specifically, the research task has proceeded through selection of candidate packaging materials, pouch configurations, and appropriate sealing parameters. Using these test materials, a series of bulk overwrap prototype systems were constructed. The systems were evaluated for feasibility, ease of stowage, and mass and volume savings over the current ISS stowage system. Oxygen scavenging systems and resealable | | |

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| | closures were also considered for their potential to enhance the bulk overwrap system further. Many of the evaluated concepts were projected to provide mass and volume savings over comparable features of the current ISS provisioning system. The use of foil materials, gusseted pouches, and oxygen scavenger systems was found to be most successful, and is therefore advised for NASA applications. Further evaluation of resealable closures may be necessary to enhance the system for application to long duration missions. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The study evaluates the feasibility of bulk packaging food items for preservation and containment of food. All prototypes considered utilize commercially available packaging materials. The results of the evaluation could be helpful to food industry. No innovative technologies are being developed. |
| **Task Progress:** | The current food system for short-duration Shuttle and International Space Station (ISS) missions utilizes prepackaged dry beverages and an assortment of shelf–stable foods. Many of these foods are packaged in primary packaging as well as a secondary overwrap pouch made of a high barrier, flexible, packaging material. This secondary overwrap material serves to extend the shelf life of these products by providing a protective barrier that prevents oxygen and moisture interaction with food. The Bulk Overwrap task was completed in 2010 and evaluated the feasibility of incorporating a bulk overwrapping system as an alternative to the individual system used on ISS. The system would provide a single, large overwrap that would contain and preserve one container's worth of food items within a high barrier, flexible material, similar to the individual overwrapping material.<br>In this task, various bulk overwrap design concepts were evaluated for feasibility and mass and volume savings over the current ISS stowage system. Specific comparisons included the use of bulk bags for containment, in place of rigid food stowage containers, and the use of bulk bags for food preservation, in place of individual overwrapping. In general, these designs were found to provide adequate storage and protection of food components, but should be considered for further compatibility with all parts of the provisioning process. Many of the features of bulk overwrap bags were projected to provide mass and volume savings over comparable features of the current ISS provisioning system (see photo). The use of foil materials, gussetted pouches, and oxygen scavenger systems was found to be most successful features, and are therefore advised for future NASA applications. Further application of resealable closures may be necessary to enhance the system for application to long-duration missions. |
| **Bibliography Type:** | Description: (Last Updated: 03/22/2018) |